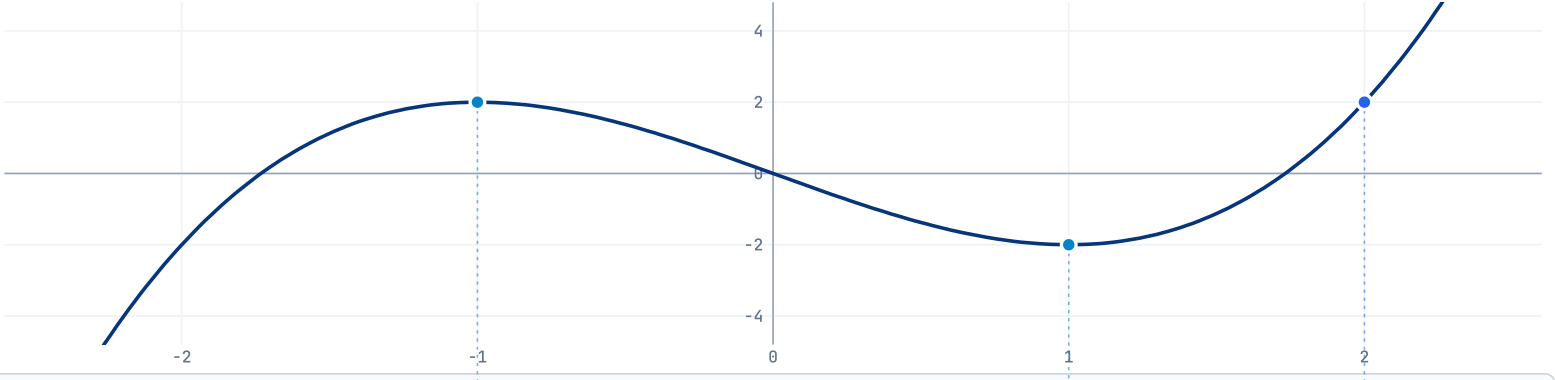
PEAKS AND VALLEYS, NEARBY ONLY

Local extrema of a cubic

$f(x) = x^3 - 3x$

1 LOCAL MAX · 1 LOCAL MIN



DIRECTION

increasing	decreasing	increasing

KEY POINTS

1 LOCAL MAXIMUM
$f(-1) = 2$

2 LOCAL MINIMUM
$f(1) = -2$

3 SAME HEIGHT
$f(2) = 2$

GLOBAL

PEAK
rise, then fall
at $x = -1$

VALLEY
fall, then rise
at $x = 1$

ABSOLUTE EXTREMA
none
the arms run off to $\pm\infty$